



D6 · DELIVERABLE

Maintenance Process

The recurring fairness audit defined as preventive maintenance: cycle, calendar, change control, KPIs and evidence.

Why a bias audit is preventive maintenance, and the recurring process that keeps HireSense fair and reliable after go-live, without depending on a one-off report.

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1 Why a fairness audit is preventive maintenance

Software maintenance is usually classified (ISO/IEC/IEEE 14764; Sommerville) as:

TYPE	PURPOSE	TRIGGER	HIRESENSE EXAMPLE
Corrective	Fix a defect after it has caused a failure	A failure is observed	BUG-04: season dates silently scored 0 years → anomaly cap added
Adaptive	Keep working in a changed environment	External change	A new OCR model or résumé tool changes text extraction
Perfective	Improve qualities or add features	User need	Intersectional audit groups; better date grammar
Preventive	Detect and remove latent faults before they become failures	Schedule, not failure	The recurring fairness audit

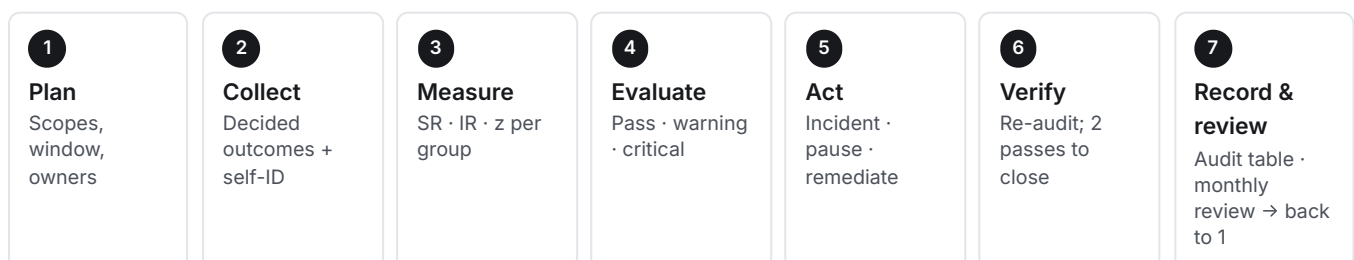
A screening system can become discriminatory **without any code change**. The applicant pool shifts, a recruiter edits a band, a new parser reads one layout better than another, or a criterion turns out to proxy a protected characteristic. These are **latent faults**: nothing crashes, and the harm accumulates silently, just like the silent parse failure. The fairness audit runs on a schedule, looks for the fault before anyone is harmed at scale, and triggers a fix. Running on a schedule to prevent failure is what makes it **preventive maintenance**.

The seeded data shows the point. HireSense never reads age, yet the audit found that the Senior Frontend role's experience bands select the 18–29 band at an impact ratio of about 0.46 (D4 §8). No test of the code could have found this, because it is a property of *criteria × applicant population*. Only recurring measurement of outcomes reveals it.

2 Sources of drift the process must catch

SOURCE	EXAMPLE	DETECTED BY
Applicant population	More early-career applicants after a campus drive	Scheduled audit (rolling 90 days)
Criteria edits	Threshold raised from 75 to 85	Preview before publish + automatic audit after
Parser / OCR release	New version reads two-column layouts differently	Evaluation gate + post-release audit
Skill taxonomy update	An alias added or removed ("JS" vs "Node.js")	Evaluation gate + audit
Proxy variables	Experience as an age proxy; location as a socio-economic proxy	Audit by every declared dimension
Process drift	Reviewers start overriding in one direction	Human-decision share and outcome KPIs

3 The recurring audit cycle



Monthly preventive-maintenance cycle (plan-do-check-act). Steps 2–4 and the containment in step 5 are automated; analysis and remediation are human.

CADENCE	ACTIVITY	OWNER	OUTPUT
Monthly (1st, 06:00)	Scheduled audit of every role + company-wide, rolling 90 days	System / audit lead	audits rows; incidents
Monthly (first week)	Governance review: open incidents, KPIs (\$6), criteria changes of the month	Audit lead + Head of Talent	Minutes, actions
Weekly (while a warning is open)	Escalated audit of the affected role	System	audits rows
Quarterly	Deep dive: threshold review ($n \geq 30$, $z \geq 1.96$), proxy analysis, self-ID non-response	Audit lead	Threshold decision record
Per release	Parser/OCR/taxonomy release gated on the evaluation corpus, then audit	Engineering	Eval report, audit
Annually	Independent review of the audit process and the tier list	External reviewer	Review report

4 Change management for criteria and parser

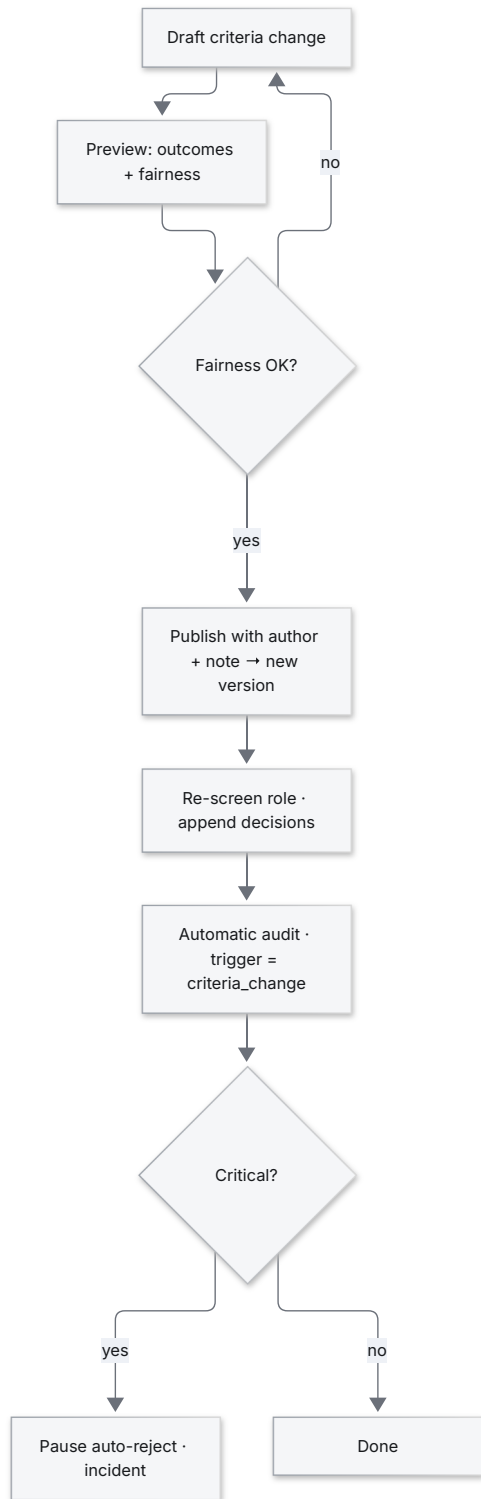


Figure 1. Criteria change workflow.

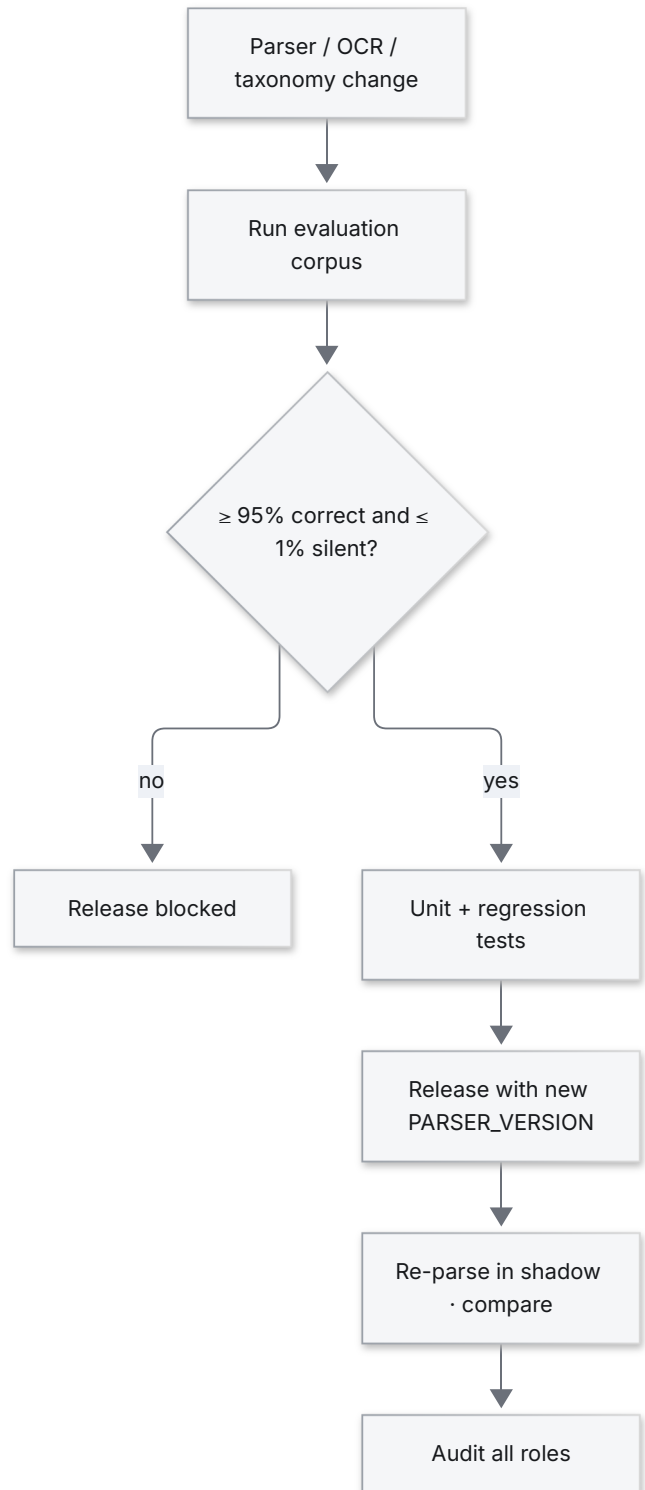


Figure 2. Parser release gate.

Every corrective fix ships with a **regression test** that would have caught the defect (D1 §8). Each release records the parser version on every parse and the criteria version on every decision, so any past decision can be explained with the exact rules and parser that produced it.

5 Event-driven (unscheduled) maintenance

EVENT	MAINTENANCE TYPE	RESPONSE
Candidate complaint or regulator query	Corrective / preventive	On-demand audit of the role; decision trace and log export within 2 working days
Critical audit	Preventive → corrective	Containment (pause) is automatic; root cause and remediation per D4 §7
New job family (e.g. non-tech roles)	Adaptive	Extend the taxonomy; evaluation corpus for the new layouts; audit after 90 days
Review queue age > 2 days	Perfective	Investigate the routing reasons; improve the parser for the dominant cause
Dependency update (OCR, model, PDF library)	Adaptive	Evaluation gate before deploy; pin versions

6 Maintenance metrics (KPIs)

SCHEDULED AUDITS 63 7 scopes × 9 months	AUDITS WITH A BREACH 27 14 critical	PARSE SUCCESS 99.0% target ≥ 95%	SILENT FAILURE RATE 0.30% target ≤ 1%
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KPI	DEFINITION	TARGET	CURRENT (SEEDED DATA)
Audit timeliness	Scheduled audits completed on the 1st ÷ due	100%	100% (seeded schedule)
Mean time to remediate (MTTR) a critical breach	Critical audit → first passing re-audit	≤ 30 days	Open: Senior Frontend (age band)
Breach rate	Audits with a breach ÷ scheduled audits	Trend ↓	43%
Parse success / silent failures	Evaluation corpus, per release	≥ 95% / ≤ 1%	99.0% / 0.30%
False-alarm routing	Correct parses sent to review ÷ all parses	≤ 1%	0.65%
Review queue	Applications awaiting a person	Age ≤ 2 days	94 waiting
Self-ID non-response	'Prefer not to say' ÷ applicants	≤ 40%	4%
Criteria change audits	Changes followed by an automatic audit	100%	100% (enforced by the API)

7 Audit history as maintenance evidence

Scheduled audits per scope (rolling 90-day windows, seeded data). Reading the history the way the monthly governance review would:

- **Persistent, real disparity.** The age band (18–29) breaches in every company-wide window and repeatedly for Senior Frontend Engineer. It comes from the experience requirement (D4 §8), and the role is paused now.
- **False alarms.** Data Analyst (university tier, two windows) and Senior Frontend (gender, one window) were graded critical, although the demo data makes tier and gender *independent* of outcomes. With about 567 group comparisons over the 9 scheduled months at the 5% level, a few chance "significant" results are expected. The pause → re-audit loop limited the cost: each cleared within one or two windows.
- **Warnings that clear on their own (W)** are the non-significant breaches. They correctly did not pause any role.

Maintenance action from this review (perfective): apply a multiple-comparison correction (Holm–Bonferroni across the groups and dimensions of a scope) before grading a breach as critical, and re-evaluate at the next quarterly deep dive. This is exactly the feedback loop a preventive-maintenance process should produce: the audit audits itself.

SCOPE	2026-01	2026-02	2026-03	2026-04	2026-05	2026-06	2026-07	2026-08	2026-09
Company-wide	C	C	C	C	C	C	C	C	C
Senior Frontend Engineer	C	✓	W	W	C	W	✓	W	C
Backend Engineer (Java)	✓	✓	✓	✓	W	W	W	W	W
Data Analyst	✓	✓	✓	W	C	✓	C	W	✓
DevOps Engineer	✓	✓	✓	✓	✓	✓	✓	✓	✓
QA Automation Engineer	✓	✓	✓	✓	✓	W	✓	✓	✓
Sales Development Representative	✓	✓	✓	✓	✓	✓	✓	W	✓

✓ pass · W warning (breach, not significant) · C critical (significant breach → auto-reject paused for a role, or escalation company-wide).

8 Maintenance log (from this project)

Corrective maintenance performed while building the prototype, classified. Each item has a regression test (D1 §8).

CHANGE	TYPE	EFFECT
Letter-spaced headings collapsed before matching	Corrective	721 standard résumés fixed for both parsers
Earliest-occurrence city match	Corrective	95 wrong cities fixed
Two-column detection: balance 8% + side-by-side check	Corrective	24 résumés fixed
Anomaly cap for unreadable sections	Preventive	New class of silent failure made visible
OCR fallback (PP-OCRv6 tiny)	Perfective	Parse success 91.1% → 99.0%; review queue 215 → 94
OCR dash/space variants in dates	Adaptive	Scanned résumés 120 → 157 of 160 correct
Grounding: employer must appear in the text	Corrective	AI assist no longer invents jobs
Significance test before pausing	Perfective	Noise-driven 'critical' audits removed
Null ratio when < 2 comparable groups	Corrective	No more vacuous 1.0 ratios